GPSDet: Geometric Parallelism Embedded Structure-Aware Slender Object Detection in Remote Sensing Images

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***Abstract*—Slender objects in remote sensing, such as bridges and trains, represent a unique, yet under-researched target because of their extreme aspect ratios. This characteristic presents challenges in effectively utilizing both low- and high-level features for detecting slender objects. In this article, we propose a geometric parallelism embedded structure-aware network for slender object detection in remote sensing. Based on the observation that slender objects often exhibit distinct parallel structures in their geometry, our method aims to extract detailed structure-aware information from low-level feature maps and inject it into the high-level box head to assist in the detection of slender objects. To capture structure-aware information of slender objects, a weakly supervised mask head and geometric parallelism regularized level set loss (GP-LS loss) are proposed. The characteristic of slender objects is modeled as the geometric parallelism regularization term to provide additional parallel shrinkage force that supervises the predicted segmentation contour closer to the slender target. In the high-level box head, the structure-aware deformable region of interest (RoI) align has been designed to learn sampling offsets from a fusion of structure-aware features and original features, effectively guiding the network to extract features that are more relevant to slender objects. Finally, when employing the sliding window detection strategy for large-size slender objects, fragmented detections may occur due to the division of objects across adjacent windows. To refine the detection bounding boxes and reduce false alarms, the fragments are processed by merging & splitting (M&S) module based on contours derived from the weakly supervised prediction mask. Experimental results on the SAT-MTB and GLH-Bridge datasets demonstrate that our approach significantly improves the slender object detection performance. The code for this work will be available on the link: https://github.com/jyzhangXdu/GPSDet**

***Index Terms*—Geometric parallelism, remote sensing, slender object detection, structure-aware deformable region of interest (RoI) align.**

Received 27 August 2025; revised 3 December 2025; accepted 5 January 2026. Date of publication 12 January 2026; date of current version 3 February 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62276197 and Grant 62171332, in part by the Fundamental Research Funds for the Central Universities under Grant QTZX25070, in part by Shaanxi Province Innovation Capability Support Plan under Grant 2023-CX-TD-09, and in part by the Postdoctoral Fellowship Program of China Postdoctoral Science Foundation (CPSF) under Grant GZC20241321 and Grant 2024M762549. *(Corresponding author: Xiangrong Zhang.)*

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Digital Object Identifier 10.1109/TGRS.2026.3653106

I. Introduction

REMOTE sensing images are typically captured from an overhead perspective using sensors mounted on satellites or aircraft. Benefiting from the dramatic increase in available remote sensing satellites, remote sensing imagery plays a crucial role in the extraction of surface information and is widely applied in fields such as urban planning [1], traffic monitoring [2], and disaster assessment [3], [4]. However, due to imaging constraints and the inherent geometric characteristics of certain objects, remote sensing images often contain a significant number of slender objects, such as roads [5], rivers [6], bridges [7], and trains [8]. As shown in Fig. 1, these objects are typically modeled and identified using distinct approaches during the interpretation process due to the significant morphological differences they exhibit.

As shown in Fig. 1(a) and (b), roads and rivers, as typical linear features, exhibit continuous spatial distribution with strong continuity and complex topological relationships. Due to their susceptibility to occlusion by buildings or vegetation, semantic segmentation methods [5], [6], [9] are more suitable for accurately delineating their shape and orientation. In contrast, as shown in Fig. 1(c) and (d), bridges and trains are isolated objects with clearly defined boundaries and limited length. Object detection methods [7], [10], [11], [12], [13] are more efficient in such cases, offering lower annotation costs and faster inference in practical applications. Given these differences, this article focuses on the task of slender object detection, aiming to improve the detection performance of targets with extreme aspect ratios, such as bridges and trains, in remote sensing imagery.

Current object detection frameworks in remote sensing, whether based on horizontal bounding box (HBB) [14], [15], [16], [17] or oriented (OBB) [18], [19], [20], [21] bounding boxes, generally fall into two architectural paradigms: single-stage and two-stage detectors. As illustrated in Fig. 2(a), two-stage detectors address large-scale variations through a hierarchical detection pipeline. In the first stage, a region proposal network (RPN) generates coarse proposals, which are then adaptively assigned to scale-specific box heads in the second stage. However, a significant challenge arises in slender object detection, particularly due to scale variation within a single target. Under existing assignment strategies, slender objects are typically allocated to the top-level detection head, determined primarily by their length. As a result, the

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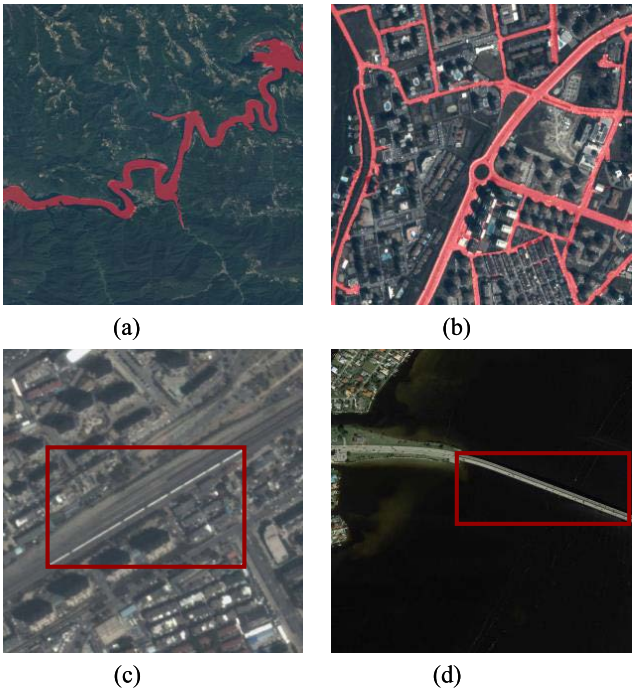


Fig. 1. Slender objects in remote sensing. (a) River. (b) Road. (c) Train. (d) Bridge.

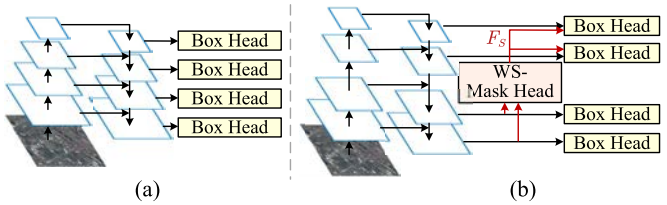


Fig. 2. Comparison between the multiscale detector and GPSDet. (a) Multiscale detector. (b) GPSDet.

high-level feature maps often have insufficient resolution to capture the fine-grained details necessary for accurate detection of slender objects. To address this issue, as shown in Fig. 2(b), we propose a weakly supervised mask head (WS-Mask Head) to extract low-level, structure-aware features. This approach not only captures the salient information of slender objects but also mitigates the misleading influence of coarse high-level features.

In this article, we focus on the task of detecting slender objects in remote sensing, specifically those with extreme aspect ratios. To address these challenges, we propose the geometric parallelism embedded structure-aware slender object detection (GPSDet) method. Our approach is based on a two-stage detector, which incorporates an additional weakly supervised mask prediction branch. This branch decouples mask prediction into dynamic kernel parameter prediction and structure-aware feature learning. It is weakly supervised by the geometric parallel regularized level set loss, utilizing only the HBB ground truth. The mask predictions are employed to refine the proposed bounding boxes generated by the RPN. To account for misalignment between sampling points and the effective features of slender objects at different levels, we designed the structure-aware deformable region of interest (RoI) align. Finally, we design a merging & splitting (M&S) processing for predicting fragments of slender objects, aiming to refine the detection bounding boxes and eliminate false alarms.

The major contributions of this article are summarized as follows.

1) A weakly supervised dynamic mask prediction branch is designed, decoupling mask prediction into dynamic kernel parameter prediction and low-level structure-aware feature learning with a centric position encoder to provide spatial information.
2) Geometric parallelism regularized level set loss (GP-LS loss) is introduced to weakly supervise the mask prediction branch using only HBB ground truth. The geometric parallelism regularized term enables the contour to shrink uniformly in both transverse and longitudinal directions.
3) The structure-aware deformable RoI align (SA-DeRoI) is modified to learn sampling offsets from the fused features rather than the original features to extract features more relevant to slender objects.
4) M&S for prediction fragments is designed to refine detection bounding boxes and eliminate false alarms of slender objects.

II. Related Work

A. Slender Object Definition

Slender objects are ubiquitous and significant in the real world, yet they have been largely neglected by previous object detection research. A slender object in the natural computer vision is defined as an object with an aspect ratio greater than 3, and the extreme slender object is defined as one with an aspect ratio greater than 5 [22]. Common slender objects in natural images include skateboards, baseball bats, pens, and so on. The effectiveness of classical object detection methods for detecting slender objects is compared and discussed in [22]. The experiments demonstrate that anchor-based localization generally outperforms anchor-free detectors for slender object detection. Moreover, specially designed box representations [10] can enhance the performance of slender object detection.

In remote sensing, the typical slender objects are bridges and trains. Bridges, as a crucial component of infrastructure, are even more ubiquitous. A large-scale bridge detection dataset named GLH-Bridge is proposed [7], which images encompass a wide range of sizes, and these bridges span diverse backgrounds and diverse scales. In the benchmark, a set of pixel intervals is defined as {(0, 50], (50, 200], (200, 800], (800, 16384]} to categorize bridges based on their lengths, classifying them as short, middle, large, and huge. SAT-MTB [8] is the first object detection dataset that fully annotates training objects in the field of view. During the preprocessing of training and testing data, slender objects are frequently cut off, which subsequently results in the misalignment of the supervision signal and the loss of contextual information.

B. Multiscale Object Detection in Remote Sensing

Due to the significant scale variation within and between object classes, multiscale object detection remains a challenging yet essential task. Convolutional neural networks (CNNs)

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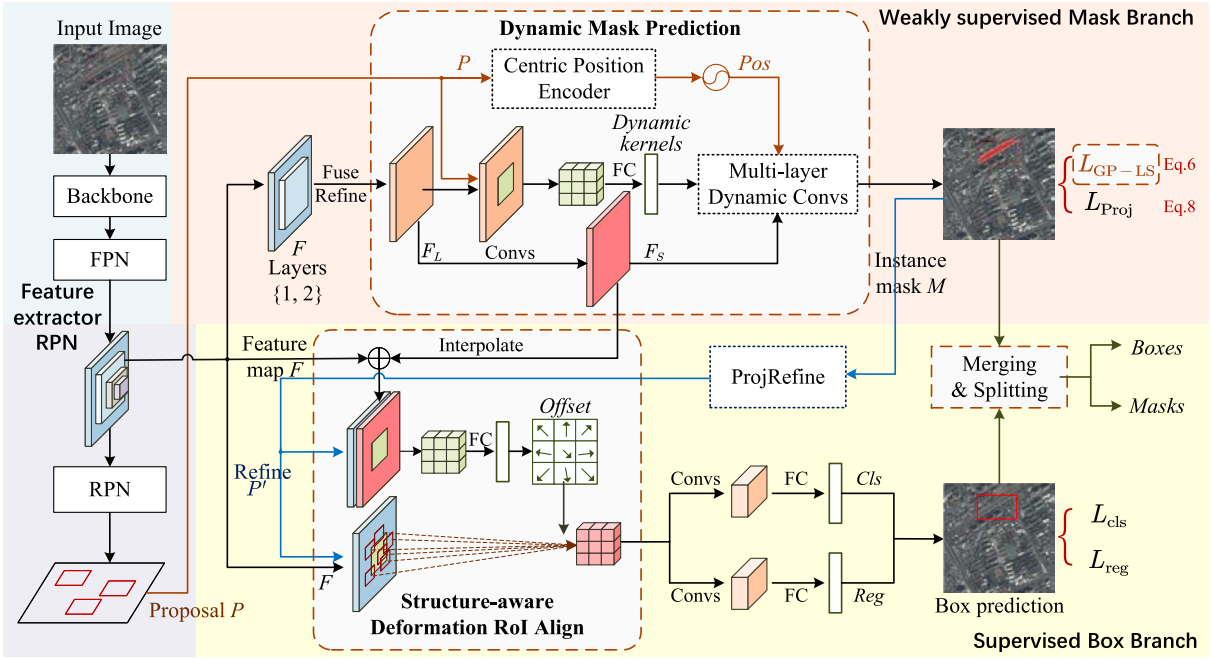


Fig. 3. Geometric parallelism embedded structure-aware slender object detection pipeline. **Black** connecting lines depict feature transfer paths, **orange** lines indicate the delivery path of the proposal bounding boxes, **blue** lines signify the transfer direction of the refined proposal bounding boxes by the mask branch, and the process indicated by the **green** line is only executed during testing.

typically employ a deep hierarchical structure. Shallow-level features often encompass structure-aware and local information, whereas high-level features possess stronger semantic information crucial for classification discrimination. To combine information from various layers, the feature pyramid network (FPN) and its derivatives are commonly utilized, facilitating the transfer of information across different layers. The feature balancing and refinement network [23] was introduced, leveraging an attention-guided balanced pyramid to semantically balance multilevel features. Additionally, the Transformer structure excels in modeling long-range dependencies.

CGL [24] employs a multiscale feature fusion module to share information between the global and local branches. The global branch can capture complete contextual information that is not split for object detection in large, very-high-resolution aerial images. GLSAN [11] proposes a downsampling strategy and self-adaptive cropping for finer-scale feature extraction.

C. Rotated Object Detection for the Extreme Aspect Ratio

Rotated object detection can predict more tightly fitting bounding boxes that better conform to object boundaries, thereby reducing background interference within predicted regions. This approach is particularly advantageous for detecting objects with extreme aspect ratios.

Several methods, Oriented R-CNN [18], Oriented RepPoints-OBB [20], and RoI Transformer [25] have investigated improved representations for rotated boxes. These studies explore various techniques, such as angle prediction, vertex sliding, or point set representations, to transform HBB predictions into OBB predictions. Other innovations include Gliding Vertex [26] and KLD [27], which propose more suitable label assignment strategies specifically adapted for OBB. Methods like Oriented R-CNN [18] and RoI Transformer [25] have designed novel alignment mechanisms to mitigate feature misalignment issues in rotated box detection. Additionally, R3Det [28], Rotated FCOS [14], and MSC-TF [29] employ feature refinement and fusion across different levels to enhance detection capability for small objects.

While OBB detection improves the localization of slender objects, most existing methods still follow conventional two-stage architectures, in which slender objects are typically assigned to high-level detection heads based on proposal size. However, the low resolution of high-level features often fails to capture the fine-grained details necessary for accurate detection. Furthermore, many widely used remote sensing datasets provide only HBB annotations, such as the SAT-MTB dataset, making the enhancement of HBB-based detection for slender objects both practical and necessary.

III. Method

A. Overview

In this section, we introduce the GPSDet for slender object detection in remote sensing. As illustrated in Fig. 3, the overall approach adheres to a two-stage detector framework. The feature extractor produces multilevel features, which are then utilized by the RPN to predict proposal bounding boxes potentially containing objects within the scene. In the subsequent stage, a weakly supervised mask head is designed to extract low-level structure-aware features F_s. Drawing inspiration from Cascade R-CNN [17], the initial proposal bounding boxes are refined by the ProjRefine module for slender objects. The refined ones are utilized to replace the

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original proposal boxes for subsequent RoI feature sampling. In the supervised box branch, the SA-DeRoI align mitigates misalignment between sampling points and slender objects across various levels. Notably, the structure-aware features F_s integrated with original features F from each level are solely employed to compute the sampling offsets for deformable RoI align. The actual features sampled remain the original features from each respective level, thereby preventing cross-level feature misguidance. Finally, in the testing process, prediction fragments are refined by the M&S module to eliminate false alarms.

B. Weakly Supervised Dynamic Mask Prediction

Drawing inspiration from Box2Mask [30], we aim to harness level set evolution to capture object structure information solely with box annotations. However, in practice, the single-stage detector employed by Box2Mask struggles to accommodate the scale variations of slender objects. Consequently, we decouple the mask prediction task into two subtasks: dynamic kernel parameter prediction and structure-aware feature learning. Additionally, we incorporate a centric position encoder to acquire the instance position prior, enhancing the model's ability to accurately locate and delineate slender objects.

1) Dynamic Kernel Parameter Prediction: As shown in Fig. 3, the predictor generates dynamic convolution kernel parameters for each proposal box. The weakly supervised mask branch is designed to extract detailed information; therefore, it takes the layer 1 and 2 features from the FPN output as input. These features are processed through a fusion module to obtain the refined feature representation F_L. Subsequently, RoI align is applied to sample the regional features of each instance based on the corresponding proposal boxes. Thereafter, the regional features are mapped to the same dimensionality as the number of multilayer convolution kernel parameters through a combination of convolutional and fully connected layers. However, the learned kernel parameters cannot perceive spatial position information. To address this, a centric position encoder is applied to produce unique position encodings for each kernel.

2) Centric Position Encoder: The centric position encoder employs a target-centric relative position encoding. First, the grid coordinates of each pixel position of the image are generated

$$\begin{aligned}[X, Y] &= \text{meshgrid}(x, y)\\ x &= 1, 2, \ldots, W\\ y &= 1, 2, \ldots, H\end{aligned} \tag{1}$$

where W and H are the width and height of the original image, respectively.

The subsequent step is to calculate the relative position offset based on the center coordinate of each proposal box. In practice, to accommodate the varying sizes of feature maps at different levels, the relative position offset is scaled to align with the feature map scale of the proposal box

$$\text{Pos} = \frac{1}{s \cdot r} \cdot \left(\left[x_c, y_c\right] - [X, Y]\right) \tag{2}$$

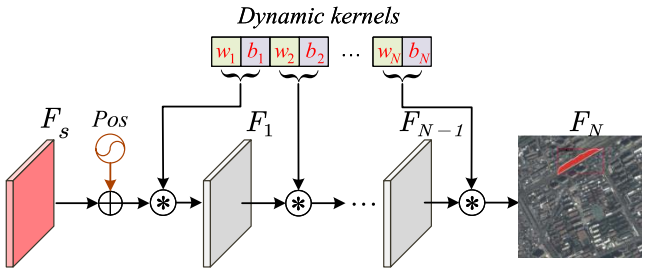


Fig. 4. Multilayer dynamic convolution.

where $[x_c, y_c]$ represents the center coordinate of the proposal box, s is the same as the downsampling stride of the corresponding feature map, and r is the size of the region, being 8.

3) Multilayer Dynamic Convolution: In the multilayer dynamic convolution module, the dynamic kernel parameters $[(w_1, b_1), (w_2, b_2), \ldots, (w_N, b_N)]$, centric position embedding Pos, and structure-aware features F_s undergo dynamic convolution in multiple stages.

The output of the final stage is the mask prediction. The structure of this module is depicted in Fig. 4. Initially, the dynamic kernel parameters are divided into N groups, serving as the convolution kernel parameters for the different layers. Each set of convolution kernel parameters comprises weights and biases. The centric position embedding Pos and structure-aware feature map F_s are concatenated as F_0, and then dynamic convolution is performed N times consecutively. Each dynamic convolution is calculated as follows:

$$F_i = \omega_i * F_{i-1} + b_i, i = 1, \ldots, N \tag{3}$$

where $\omega_i \in R^{C \times k \times k}$ is the weight and b_i is the bias of the i_{th} convolution.

4) Geometric Parallelism Regularized Level Set Loss: The level set evolution is a well-established variational method for image segmentation [31]. It represents the implicit curve through an energy function in a higher-dimensional space and iteratively minimizes this energy to refine the contour. In the classical continuous Chan-Vese energy-based level set model, the energy function is defined as follows:

$$\begin{aligned}L_{\text{LS}} &= \mathcal{F}^{\text{CV}}\\ &= \int_{\Omega} |I(x,y) - c_1|^2 \, H(\phi(x,y)) dxdy\\ &\quad + \int_{\Omega} |I(x,y) - c_2|^2 \, (1 - H(\phi(x,y))) dxdy\\ &\quad + \gamma \int_{\Omega} |\nabla H(\phi(x,y))| \, dxdy\end{aligned} \tag{4}$$

where H is the Heaviside function and $\phi(x, y)$ is the level set function. I represents the given image and Ω is the initial region. A parametric contour $C = \{(x, y) : \phi(x, y) = 0\}$ chops the region up into inside contour C: $\Omega_1 = \{(x, y) : \phi(x, y) > 0\}$ and outside contour C: $\Omega_2 = \{(x, y) : \phi(x, y) < 0\}$. c_1 and c_2 are the mean values of input $I(x, y)$ inside C and outside C. In the energy function $\mathcal{F}^{\text{CV}}$, the first two terms intend to fit the data. And the third term is actually equivalent to the length of the contour, driving the contour to shrink inward from the initial region edge.

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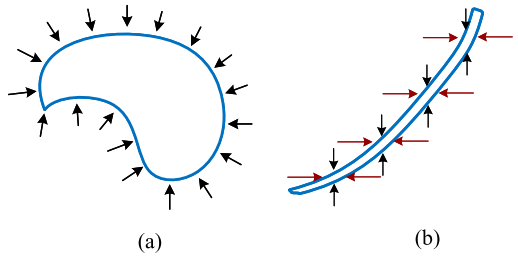


Fig. 5. Illustration of the regularized term driving. (a) Length regularized term. (b) Geometric parallelism regularized term.

Fig. 5(a) demonstrates the impact of the length regularized term on the level set function. Since objects in natural images often exhibit irregular shapes, the length regularized term typically causes the contour to shrink uniformly and inward. However, slender objects in remote sensing imagery, such as trains and bridges, typically possess regular contours with geometric parallelism both vertically and horizontally. To address this, we have introduced an additional geometric parallelism regularized term to the level set energy function, as follows:

$$\begin{aligned} R_{\mathrm{GP}} &= \beta \cdot \frac{1}{2} \cdot \frac{1}{W_\Omega} \left(\frac{1}{H_\Omega} \sum_{i=1}^{H_\Omega} \left| w_i - w_{\mathrm{avg}} \right| \right) \\ &\quad + \beta \cdot \frac{1}{2} \cdot \frac{1}{H_\Omega} \left(\frac{1}{W_\Omega} \sum_{j=1}^{W_\Omega} \left| h_j - h_{\mathrm{avg}} \right| \right) \\ &= \frac{\beta}{2 \cdot S_\Omega} \cdot \left(\sum_{i=1}^{H_\Omega} \left| w_i - w_{\mathrm{avg}} \right| + \sum_{j=1}^{W_\Omega} \left| h_j - h_{\mathrm{avg}} \right| \right). \end{aligned} \tag{5}$$

Since the computation is performed on the image, we utilize a discrete representation. In the formula (5), H_Ω, W_Ω, and S_Ω represent the height, width, and area of the minimal outer rectangle enclosing the contour, respectively. The first term calculates the sum of absolute differences between the width of each row and the average width, and this sum is then normalized by dividing by the width W_Ω. Similarly, the second term is applied vertically. As shown in Fig. 5(b), R_{GP} encourages the contour to shrink uniformly in both the horizontal and vertical directions.

This term is incorporated into the level set energy function as an optimization function for the weakly supervised mask prediction branch

$$L_{\mathrm{GP-LS}} = L_{\mathrm{LS}} + \lambda \cdot R_{\mathrm{GP}} \tag{6}$$

where the coefficient λ serves as the regularization term's weight.

5) Mask Projection: Mask projection refers to the process of projecting the mask onto both horizontal and vertical directions. The results of the mask projection will be utilized as supervision for the mask branch and to refine the proposal boxes in the ProjRefine module.

The predicted mask score is noted as $m \in [0,1]^{H \times W}$. The horizontal and vertical projection operator is defined as follows:

$$\begin{aligned} \mathrm{Proj}_x(m) &= \max_y(m) = l_x^m \\ \mathrm{Proj}_y(m) &= \max_x(m) = l_y^m \end{aligned} \tag{7}$$

where $\mathrm{Proj}_x : R^{H \times W} \to R^W$ and $\mathrm{Proj}_y : R^{H \times W} \to R^H$ indicate that projecting the mask onto x-axis and y-axis, respectively. $l_x \in [0,1]^W$ denotes the 1-D segmentation mask on the x-axis and the same applies to l_y.

The projection loss is calculated between the projections of the ground-truth box and the predicted mask as follows:

$$L_{\mathrm{Proj}} = L_{\mathrm{Dice}}(l_x^m, l_x^b) + L_{\mathrm{Dice}}(l_y^m, l_y^b) \tag{8}$$

where b is the ground-truth box and $L_{\mathrm{Dice}}(\cdot,\cdot)$ is the Dice loss [32]. $L_{\mathrm{GP-LS}}$ and L_{Proj} collaboratively provide weak supervision to the mask prediction branch, utilizing solely horizontal box annotations.

C. Structure-Aware Deformation RoI Align

Previous studies [22] have demonstrated that deformable convolution enhances the detection of slender objects. In remote sensing imagery, both large and small targets are present. To maintain the detection performance of small objects, we have designed SA-DeRoI. Compared to the traditional DeRoI, SA-DeRoI incorporates structure-aware features to learn sampling offsets that are aligned with slender objects. This method not only captures the structure-aware information of slender targets but also avoids the loss of global information in high-level features.

As illustrated in Fig. 3, the input F_c consists of structure-aware features F_s and original features F from the feature pyramid

$$F_c = \mathrm{concate}(F, F_s). \tag{9}$$

The fused feature map F_c is utilized to learn the sampling offset $\Delta p \in \mathbb{R}^{H \times W \times 2k^2}$. $2k^2$ corresponds to the 2-D offset of the $k \times k$ convolution kernel in the x- and y-directions. For each position p_0, calculate the sampling position after offset p_n

$$\Delta p = w_{\mathrm{offset}} * F_c \tag{10}$$

$$p_n = p_0 + p_n^{\mathrm{grid}} + \Delta p_n \tag{11}$$

where p_n^{grid} is the regular grid positions of the standard convolution and $n \in [0, 2k^2]$ represents the index of the convolution position.

Then, the offset sampling position p_n guides the RoI Align to extract features that are more relevant to the slender object, thereby enhancing subsequent classification and localization tasks

$$f(p_n) = \sum_q G(q, p_n) \cdot f(q) \tag{12}$$

where $G(\cdot)$ is the bilinear interpolation kernel and q traverses the four nearest integer grid points of p_n.

The actual features sampled remain the original features F from each respective level, thereby preventing cross-level feature misguidance.

D. M&S for Fragments

In the training and testing of common detections, the original scenes are processed using an overlapping patches-based cropping strategy. For example, the cropping window size is 1024 × 1024 pixels with a 200-pixel overlap in the GLH-Bridge benchmark. However, the giant bridges exceeding 3000 pixels with an extreme aspect ratio are always crossing

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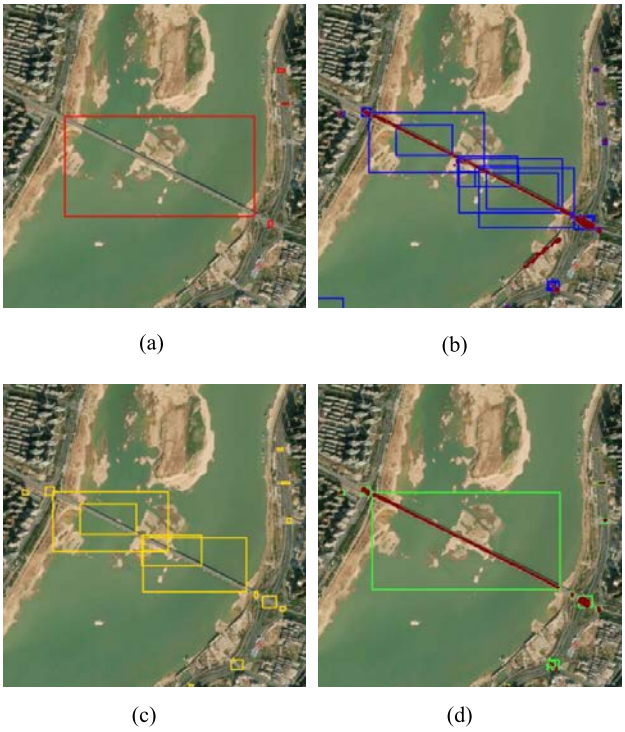


Fig. 6. Visualization of M&S for fragments. (a) Ground truth. (b) Prediction before merging. (c) Prediction after merging by NMS. (d) Prediction after M&S.

several patches. As shown in Fig. 6(b), the predicted bounding boxes contain many fragments due to the sizes of slender targets and the application of sliding window detection. The basic NMS methods can only use intersection to intersection and union (IoU) and score to screen the detection results of different patches, but cannot optimize the splicing of bridge fragments, as illustrated in Fig. 6(c). As a result, many fragments of slender objects are produced.

The weakly supervised mask branch can predict the masks corresponding to the bounding box prediction as an auxiliary tool for optimizing predicted HBB results. The basic motivation of M&S for fragments is that segmentation masks belonging to the same slender target will overlap. Merge two masks and split them according to the connected components. If they are similar to the original box, the region can be considered as the target, corresponding to the optimized HBB. Otherwise, it may belong to other adjacent targets or noise.

The pseudo-code of the M&S for fragments is shown in Algorithm 1. First, the fragment with the highest score is selected, and then the IoU is calculated between the selected fragment and the others. For all fragments with IoU greater than the threshold T, perform the process of M&S. In the process of M&S, the combined mask m_{or} of mask m_i and mask m_d is first computed through a logical OR operation. Then, based on the contours of the mask m_{or}, the mask is partitioned into distinct regions. The HBB b_k of the subregion c_k is used to calculate the $\text{IoU}_{i,k}$ and $\text{IoU}_{d,k}$ with respect to b_i and b_d, respectively. Subregions where either $\text{IoU}_{i,k}$ or $\text{IoU}_{d,k}$ exceeds the threshold T are retained. Among these, the subregion with the highest $\text{IoU}_{i,k}$ replaces the original m_i as the mask for the selected fragment in the next iteration. The iteration continues until no fragments remain. The processed result is depicted in Fig. 6(d), demonstrating that the M&S module is capable of locating slender targets with greater completeness.

Algorithm 1 M&S for Fragments

Data:
$I = \{[b_1, s_1, m_1], \ldots, [b_N, s_N, m_N]\}$, T
b is the initial detection box
s is the corresponding detection score
m is the corresponding segmentation mask
T is the IoU threshold
Result: $O = \{[b_1, s_1, m_1], \ldots, [b_o, s_o, m_o]\}$

Initial: $O \leftarrow \{\}$;
while $I \neq \emptyset$ **do**
  $i \leftarrow \textbf{argmax}(S)$;
  $D \leftarrow \{d | \text{IoU}(b_i, b_d) \geqslant 0, d \in I\}$;
  $R \leftarrow \{\}$;
  for d *in* D **do**
    $m_{or} \leftarrow m_i \vee m_d$;
    $Cons \leftarrow \textbf{findContours}(m_{or})$;
    $IoU_i^{max} \leftarrow 0$, $k^{max} \leftarrow -1$;
    for c_k *in* $Cons$ **do**
      $b_k \leftarrow \textbf{Proj}(c_k)$;
      $m_k \leftarrow \textbf{Filled}(c_k)$;
      $IoU_{i,k} \leftarrow \text{IoU}(b_i, b_k)$;
      $IoU_{d,k} \leftarrow \text{IoU}(b_d, b_k)$;
      if $\left(IoU_{i,k} \geqslant IoU_{d,k}\right) \wedge \left(IoU_{i,k} \geqslant T\right)$ **then**
        $R \leftarrow R + [b_k, s_i, m_k]$;
        if $IoU_{i,k} > IoU_i^{max}$ **then**
          $k^{max} \leftarrow k$.
      if $\left(IoU_{i,k} < IoU_{d,k}\right) \wedge \left(IoU_{\text{d},k} \geqslant T\right)$ **then**
        $R \leftarrow R + [b_k, s_d, m_k]$.
    if $k^{max} = -1$ **then**
      break
    else
      $[b_i, s_i, m_i] \leftarrow \left[b_k^{max}, s_k^{max}, m_k^{max}\right]$;
      $R \leftarrow R - [b_k^{max}, s_k^{max}, m_k^{max}]$.
  $O \leftarrow O + [b_i, s_i, m_i]$;
  $I \leftarrow I - D + R$.
return O

E. Training and Testing Pipeline

Compared with natural scenes, the ratio of almost all objects' size to scene size in remote sensing images is typically smaller. Consequently, adjustments to the parameters and pipelines for training and testing are necessary. The specific adjustments include the following:

1) Scale Selection Threshold: In RoI Align, the appropriate scale feature is chosen based on the size of the proposal box, as demonstrated in the following equation:

$$\begin{aligned} L_0 &: R_{\text{scale}} < R_{\text{scale}}^{\text{finest}} \times 2 \\ L_1 &: R_{\text{scale}}^{\text{finest}} \times 2 \leq R_{\text{scale}} < R_{\text{scale}}^{\text{finest}} \times 4 \\ L_2 &: R_{\text{scale}}^{\text{finest}} \times 4 \leq R_{\text{scale}} < R_{\text{scale}}^{\text{finest}} \times 8 \\ L_3 &: R_{\text{scale}} \leq R_{\text{scale}}^{\text{finest}} \times 8 \end{aligned} \tag{13}$$

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where $R_{\text{scale}} = (h \times w)^{1/2}$ represents the scale of a proposal box. $R_{\text{scale}}^{\text{finest}}$ is the parameter of the scale selection threshold.

In natural object detection, such as when Faster R-CNN [16] is employed on the MSCOCO dataset [33], the parameter $R_{\text{scale}}^{\text{finest}}$ is typically set to 56. However, in remote sensing, the sizes of different targets can vary significantly. To analyze this, we counted the distribution of instances in the SAT-MTB dataset at the feature level when $R_{\text{scale}}^{\text{finest}}$ was set to 56.

As illustrated in Fig. 7(a), tiny objects, such as ships, are concentrated in the L_0 feature level, while large-sized objects constitute only a small proportion. Consequently, the high-level detection branch may not receive adequate training. Additionally, the unbalanced scale distribution can cause the RPN to generate a disproportionate number of small proposal boxes. Therefore, we set $R_{\text{scale}}^{\text{finest}} = 28$ in our framework, and the basic size of the anchor in RPN is changed from 8 to 4. Fig. 7(b) shows the feature-level distribution of instances when $R_{\text{scale}}^{\text{finest}} = 28$.

It is evident that training instances are primarily concentrated in higher feature levels to capture richer global information. Additionally, small instances exhibit a more balanced distribution across different feature levels.

2) RPN Label Assignment: To address the issue of significant variation in instance size, a hybrid label assignment strategy is employed during RPN training. Given that IoU is highly sensitive to the positional deviations of tiny objects, the normal Gaussian Wasserstein distance (NWD) [34] is utilized for the label assignment of small objects. Compared to the traditional IoU-based label assignment strategy, NWD provides smoother handling of positional deviations and the ability to measure the similarity between nonoverlapping or mutually inclusive bounding boxes. For large objects, the same IoU-based label assignment strategy as in Faster R-CNN [16] is retained. The criteria for dividing large and small objects are consistent with the L_0 scale selection, where ground-truth bounding boxes with $R_{\text{scale}} < R_{\text{scale}}^{\text{finest}} \times 2$ are considered to be small objects.

3) Large Proposal Box Refined: To prevent error accumulation, the bounding box refinement through the ProjRefine operation is only applied to large proposal boxes. The criterion for identifying large proposal boxes is that their size $R_{\text{scale}}^{\text{large}} = R_{\text{scale}}^{\text{finest}} \times 4$. In other words, the proposal boxes belonging to levels L_2 and L_3 will undergo refinement through the ProjRefine operation.

IV. Experiments

A. Dataset Description

1) SAT-MTB Dataset: The SAT-MTB dataset [8] is an optical remote sensing object detection dataset. The dataset contains 144 video sequences captured by the Jilin-1 satellite, for a total of 308 204 annotated instances. The spatial resolution is 0.92 m per pixel. The categories of interest encompass airplanes, ships, and trains. Notably, these different categories of objects exhibit significant diversity in terms of their spatial distribution, size, orientation, and other attributes.

However, the limited number of train instances presents significant challenges for thoroughly training and validating detection models. To mitigate this issue, we have expanded the SAT-MTB dataset as the SAT-MTB+ dataset. SatSOT [35], a dataset designed for satellite video single object tracking, contains 25 sequences featuring train objects. Importantly, these videos were captured by the same satellite, Jilin-1, as the images in SAT-MTB, thereby eliminating concerns about data distribution discrepancies. The statistical comparison before and after data expansion is shown in Table I. We have removed duplicate scenes and reannotated the objects of interest to create an augmented dataset. The annotation tool utilized in this study is derived from SV248S [36]. The primary function of this tool is to mark the key frames of the target and interpolate the intermediate frames. In comparison to the Labelme [37] tool, which is commonly used in the construction of detection datasets, the annotation results obtained using this tool exhibit greater consistency.

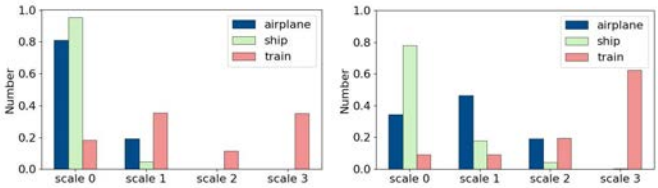


Fig. 7. Scale distribution in the SAT-MTB dataset. (a) Finest scale 56. (b) Finest scale 28.

TABLE I
Statistical Comparison Between the SAT-MTB and SAT-MTB+ Dataset

	SAT-MTB		SAT-MTB+	
	Videos	Instances	Videos	Instances
Airplane	62	193,849	62	194,809(+0.5%)
Ship	70	81,734	70	81,734
Train	10	6,795	28(+180%)	11,762(+73.1%)

2) GLH-Bridge Dataset: GLH-Bridge [7] is a large-scale bridge detection dataset for remote sensing, which contains 6000 VHR RSIs sampled from diverse geographic locations across the globe. These images encompass a wide range of sizes, varying from 2048 × 2048 to 16 384 × 16 384 pixels, and collectively feature 59 737 bridges. The spatial resolution varies from 0.3 to 1.0 m. In contrast to the other previous bridge detection datasets, GLH-Bridge incorporates a diverse range of bridge sizes, ranging from tiny bridges with 12 pixels to giant bridges exceeding 3000 pixels. Large bridges show a high presence in GLH-Bridge with extreme aspect ratios.

B. Evaluation Metrics

For evaluation metrics, we adopt the mean average precision (mAP) calculation, as used in the PASCAL VOC07 dataset [38]. First, the prediction boxes are arranged from high to low according to their confidence scores, and then the IoU with the annotation boxes. Once a ground truth box is correctly matched to a prediction box, it is no longer considered for matching with other prediction boxes. A prediction box is considered a true positive (TP) if its IoU with the ground-truth box is greater than the threshold. Subsequently, we calculate

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the precision and recall for the previously matched boxes

$$\text{Precision} = \frac{\text{TP}}{N_{\text{Pred}}}$$
$$\text{Recall} = \frac{\text{TP}}{N_{\text{Anno}}} \tag{14}$$

where N_{Pred} and N_{Anno} represent the number of prediction boxes and annotation boxes, respectively. To form a precision–recall (P–R) curve, we traverse all prediction boxes. The area under the P–R curve represents the AP value for a given category. To evaluate in accordance with the PASCAL VOC07 evaluation, we calculate the precision at each point where the recall ranges from 0.0 to 1.0 in steps of 0.1.

As a benchmark for the SAT-MTB dataset, the IoU threshold for TP is set to 0.5. The AP for each category of objects is obtained by summing and averaging these precision values. The mean of the AP values for all categories is the mAP value.

For better assess the efficacy of these methods in GLH-Bridge benchmark, a set of pixel intervals are defined as [0, 50], (50, 200], (200, 800], (800, ∞) to categorize bridges based on their average side length of HBB, classifying them as short, middle, large, and huge. The corresponding APs are denoted as AP_{sh}, AP_{md}, AP_{lg}, and AP_{hg}, respectively. AP_{50} and AP_{75} set the IoU threshold for TP to 0.5 and 0.75, respectively. $AP_{50:95}$ is the AP averaged over IoU thresholds ranging from 0.5 to 0.95 for TP.

C. Implementation Details

Our method and the comparison algorithm are based on the MMDetection framework [39] and implemented in PyTorch 1.8.0 on a computer equipped with 4 V100 GPUs with 32 GB of memory, a 2.20 GHz Intel Xeon CPU, and 64 GB of RAM.

The training and test data preprocessing follows the method outlined in the DOTA dataset [40]. For SAT-MTB datasets, the images are divided into 800×800 patches with an interval of 270 and a scaling ratio of 1. Both our method and the ablation detector were trained using the AdamW optimizer for 8 epochs, with a batch size of 16, a learning rate of $1e-4$, and a weight decay of $1e^{-4}$. No multiscale data augmentation was used. The training takes 17.25 h.

For the GLH-Bridge dataset, the training settings are consistent with the GLH-Bridge benchmark [7]. The images are divided into 1024×1024 patches with an interval of 200 and a scaling ratio of 1. A total of 24 epochs were trained. The training time is approximately 22 h.

D. Evaluation on the SAT-MTB Dataset

We train and evaluate each method on the SAT-MTB datasets [8]. The training parameter settings for the contrastive method are identical to those used in the SAT-MTB benchmark [8].

Experimental results on the SAT-MTB datasets are shown in Table II. Most methods achieve good performance in aircraft category object detection. Faster R-CNN [16], which serves as the baseline for our method, only achieves an mAP score of 24.0% on the train category. From the visualization results in Fig. 8, it can also be observed that the Faster R-CNN detector misses a significant portion of the train targets. In contrast, Cascade R-CNN [17] adopts a multistage optimization approach from low to high quality, achieving better performance. The performance of FCOS [14] and ATSS [15] on the SAT-MTB dataset is moderate, primarily attributed to their inability to effectively tackle the challenges posed by diverse target scale variations and sample imbalance. RepPoints [41] utilizes a set of sample points to represent objects, offering a more refined granularity for object localization compared to traditional bounding box representations. However, this innovative approach also presents a challenge, specifically the generation of a substantial number of false positive detections. As shown in Fig. 8, DETR [42] delivers relatively low performance at detecting small objects. For CNN-based detectors, small objects are often detected from high-resolution feature maps. However, high-resolution feature maps lead to unacceptable complexities for DETR.

TABLE II
EXPERIMENTS ON THE SAT-MTB DATASET

	Airplane	Ship	Train	mAP(%)	Params(M)	FPS
Faster R-CNN	75.7	37.4	35.4	49.5	41.2	19.0
Cascade R-CNN	**77.0**	33.0	40.8	50.3	68.9	15.7
FCOS	75.7	32.8	31.8	46.8	**31.2**	15.9
ATSS	74.7	30.3	33.9	46.3	31.9	**21.2**
RepPoints	74.5	37.0	27.1	46.2	36.3	20.2
DETR	47.6	31.2	35.3	38.1	41.2	7.9
Our	76.6	**38.6**	**61.2**	**58.8**	158.2	5.6

TABLE III
EXPERIMENTS ON THE SAT-MTB+ DATASET

	Airplane	Ship	Train	mAP(%)
Faster RCNN	75.3	25.5	48.9	49.9
Cascade RCNN	76.2	34.1	55.2	55.2
FCOS	**77.0**	28.1	50.1	51.7
ATSS	75.9	33.7	48.0	52.5
RepPoints	76.2	39.5	57.1	57.6
DETR	46.9	25.9	52.7	41.8
Our	75.8	**41.6**	**64.8**	**60.7**

In contrast, our method achieves substantial improvements in the mAP (+9.3%), particularly in the train (+20.4%) and ship (+1.2%) categories, by exploring the properties of geometric parallelism and integrating structure-aware features.

Table II also shows the parameter size and inference speed of each method. Our method requires additional computation for the mask, which results in slightly slower inference speed. It is worth noting that the parameter size of the structure-aware deformation RoI Align module is 114.0 MB. For applications with limited computational resources, this module can be disabled to strike a balance between performance and efficiency.

To investigate the influence of the augmented dataset, we train and evaluate each method using the SAT-MTB+ dataset. As shown in Table III, the SAT-MTB+ dataset's richer and more diverse train samples significantly enhance the object detection performance of these methods for the train category. The experimental results indicate that Cascade R-CNN [17] and RepPoints [41] exhibit competitive performance compared to previous methods. Our proposed method achieves the best performance in detecting trains, ships, and in the overall evaluation.

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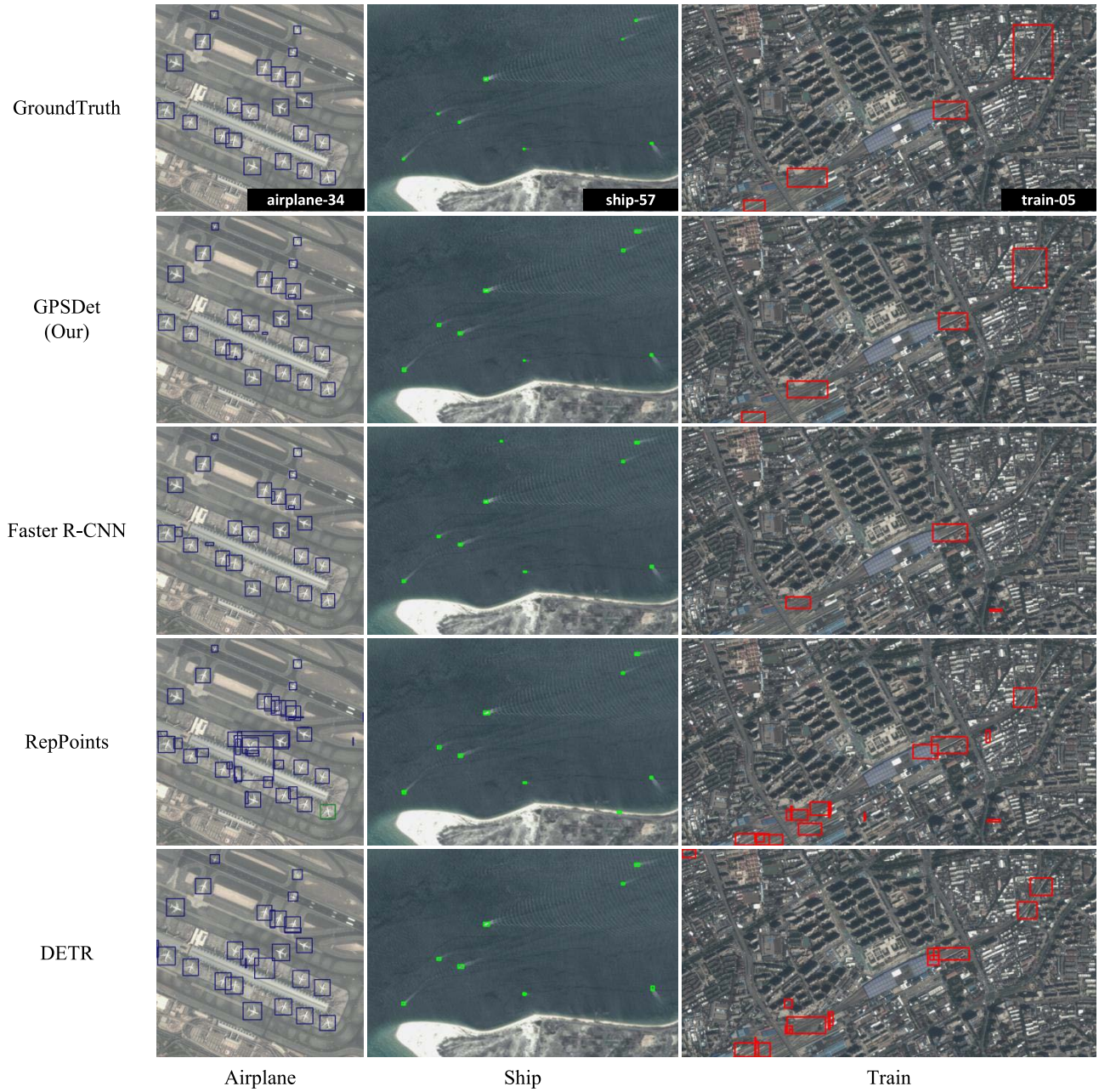


Fig. 8. Visualization of results on the SAT-MTB dataset.

TABLE IV
EXPERIMENTS ON THE GLH-BRIDGE DATASET

	$AP_{50:95}$	AP_{50}	AP_{75}	AP_{sh}	AP_{md}	AP_{lg}	AP_{hg}
Faster R-CNN	32.2	61.5	26.5	27.3	36.8	20.4	4.5
Cascade R-CNN	30.6	60.8	27.3	27.2	35.5	19.7	3.3
FCOS	25.4	51.6	22.5	22.1	30.8	19.7	5.2
ATSS	27.6	54.2	23.8	24.4	33.1	22.4	5.8
RepPoints	28.4	55.3	24.3	25.4	34.5	21.8	5.0
GLSAN	31.6	58.5	26.1	27.1	35.5	19.7	15.6
SAHI	32.1	61.2	26.5	26.8	**37.1**	21.3	21.0
Our	**32.4**	**64.0**	**28.0**	**28.9**	37.0	**30.3**	**44.6**

E. Evaluation on the GLH-Bridge Dataset

Experimental results on the GLH-Bridge datasets are shown in Table IV. GLH-Bridge [7] is a large-scale, very high-resolution bridge detection dataset, which incorporates a diverse range of bridge sizes, ranging from tiny bridges with 12 pixels to giant bridges exceeding 3000 pixels. From the performance evaluation of all methods, it can be seen that common detectors have much lower accuracy in detecting large bridges than in detecting medium-sized ones. The detection of huge bridges can often be a challenging task, as they may not be effectively and completely captured within a single sliding window when employing traditional cropping strategies. Here, we include two additional detection methods, GLSAN [11] and SAHI [12], specifically designed for large-scene and large-object targets for comparative analysis. GLSAN [11] is a global–local detection network with an efficient self-adaptive region selection. In the experiment, the local region size is set to 1024 with the global region size being 2×. SAHI [12] provides a generic slicing-aided inference and overlapping patches-based cropping strategy for various objects. For

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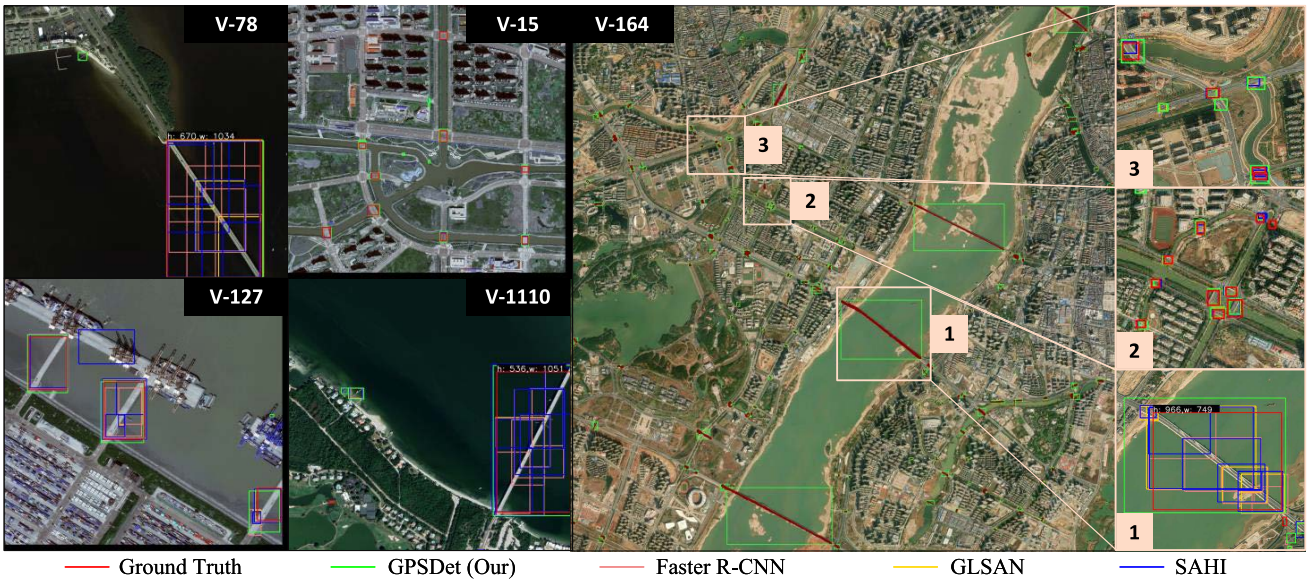


Fig. 9. Visualization of results on the GLH-Bridge dataset.

TABLE V
COMPREHENSIVE ABLATION OF KEY MODULES FOR DETECTION PERFORMANCE

	WS-Mask Head	SA-DeRoI	M&S	Aircraft	Ship	Train	mAP (%)	Params (M)
(a)				75.3	25.5	48.9	49.9	**41.2**
(b)	✓			75.2	38.4	50.8	54.8	44.3
(c)	✓	✓		**76.6**	**38.6**	60.3	58.5	**158.2**
(d)	✓		✓	75.2	38.4	53.6	55.7	44.3
(e)	✓	✓	✓	**76.6**	**38.6**	**61.2**	**58.8**	**158.2**

TABLE VI
ABLATION EXPERIMENTS FOR L_{LS} AND L_{GP-LS}

LS loss	GP-LS loss	Aircraft	Ship	Train	mAP(%)
✓		75.1	**39.6**	50.5	55.0
	✓	**76.6**	38.6	**61.2**	**58.8**

TABLE VII
ABLATION EXPERIMENTS FOR SA-DEROI

	F	F_s	DeRoI	SA-DeRoI	Aircraft	Ship	Train	mAP(%)	Params(M)
(a)	✓				75.2	38.4	53.6	55.7	**44.3**
(b)	✓		✓		75.2	38.4	49.8	54.1	151.8
(c)		✓			74.1	**43.4**	50.2	56.2	45.1
(d)		✓	✓		76.1	41.3	37.2	51.2	**159.1**
(e)		✓		✓	**76.6**	38.6	**61.2**	**58.8**	158.2

fairness, slicing-aided fine-tuning is not enabled. As evident from the experimental results Table IV, the combination of global–local search, along with efficient regional selection and cropping strategies, contributes to enhancing the performance of large-scale and huge-scale bridge object detection.

Our method has achieved significant improvement in the GLH-Bridge dataset. From Table IV and Fig. 9, it can be seen that our method achieves significant improvement (+23.6%) in the detection of huge bridges only with a single, uniformly weighted detector. M&S for fragments utilizes masks corresponding to prediction boxes to fuse, classify, and discriminate target masks, thereby optimizing horizontal box prediction.

F. Ablation Experiments

The ablation experiments are divided into four parts, each exploring the impact of: 1) the geometric parallelism regularization term L_{GP-LS}; 2) SA-DeRoI; 3) the ProjRefine module and M&S for fragments; and 4) feature fusion strategies in the SA-DeRoI module.

Table V summarizes the overall ablation results when progressively integrating these modules. As shown in Table V, compared with the baseline in (a), adding the weakly supervised mask head already improves the mAP from 49.9% to 54.8%, indicating that learning detailed structure features benefits slender object detection. Further introducing the SA-DeRoI with regularization (c) increases the mAP to 58.5%, showing clear gains without relying on M&S. By contrast, applying M&S alone (d) yields a lower mAP (55.7%) than the regularization approach in (c), suggesting that the improvement is not merely due to merging operations but also from the proposed SA-DeRoI and regularization which enhances feature representation. The best result (58.8%) in (e) is achieved when combining both strategies, indicating their complementary effect. These results align with the findings from our more focused ablation studies for each module, highlighting that the improvements stem not only from merging operations but also from the proposed structural and geometric constraints.

To isolate the impact of geometric parallelism regularization, we further conduct experiments comparing the standard level set loss with the addition of the GP-LS loss in the weakly supervised mask head. In the ablation experiment, we replace the GP-LS loss L_{GP-LS} with the original level set loss L_{LS}. As shown in Table VI, the train detection performance using L_{LS} is 10.7% lower than that using L_{GP-LS}.

In the second phase of the ablation experiment, we explore the impact of SA-DeRoI. In experiment (a) of Table VII, the box branch uses original features without deformable RoI Align, named DeRoI. In other words, the box prediction

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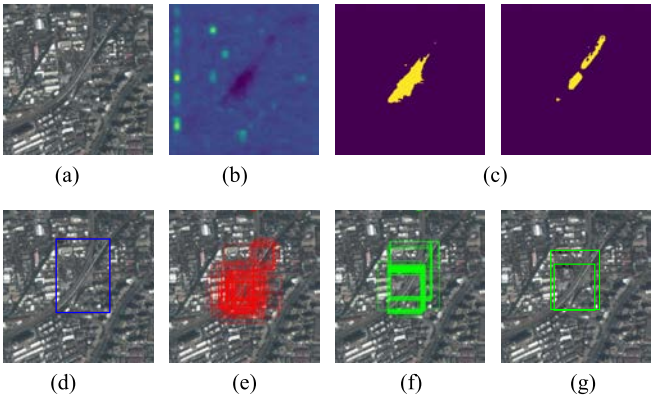


Fig. 10. Visualization of the ProjRefine module for refining proposal boxes. (a) Image. (b) Structure-aware feature. (c) Weakly supervised mask predictions. (f) Ground truth. (d) RPN proposal boxes. (e) Refined proposal boxes. (g) Final results.

branch maintains the same structure as in Faster R-CNN [16]. It can be observed that the incorporation of a weakly supervised mask branch aids the backbone in extracting features that are beneficial for training detection. In experiment (b), the box branch uses original features with deformable RoI Align. But the training detection performance actually decreases. A possible reason is that it is challenging for high-level features to effectively predict the positional offsets for DeRoI. In experiments (c) and (d), the original features F is replaced with structure-aware ones F_s. In contrast, while the detection performance for small-sized ship targets is improved, the detection performance for large-sized trains decreases. The reason is that structure-aware features, which contain more low-level detailed features, are more suitable for small object detection. In experiment (e), SA-DeRoI incorporates structure-aware features F_s only to learn sampling offsets, while still sampling the target region features from the original features F. This method not only captures the structure-aware information of slender objects but also avoids the loss of global information in high-level features, achieving the best performance.

To address concerns about model complexity, we provide a fine-grained breakdown of parameter distribution across core components in Tables V and VII. Notably, the WS-Mask head contributes minimally to the overall parameter count: adding it increases the total number of parameters by only around 3 M (e.g., from 41.2M to 44.3M in Table V(a)→(b), underscoring its lightweight design. In contrast, the dominant source of parameter growth is the deformable convolution module. When replacing standard RoIAlign with DeRoI, parameters jump from 44.3M to 151.8M [Table VII(a)→(b)], primarily due to dense offset prediction over full feature maps. Our proposed SA-DeRoI mitigates this overhead by restricting offset computation to low-dimensional structure-aware features F_s, resulting in lower parameter count [158.2M versus 159.1M in Table VII(d)→(e)] and higher accuracy (+ 7.6% mAP on training). This indicates that SA-DeRoI improves parameter efficiency by focusing deformable modeling on semantically relevant regions.

In the third phase of the ablation experiment, as shown in Table VIII, we explore the impact of ProjRefine and M&S for fragments. As shown in Fig. 10, the implementation of

TABLE VIII
ABLATION EXPERIMENTS FOR REFINE STRATEGIES

ProjRefine	M&S	Aircraft	Ship	Train	mAP(%)
		76.5	**41.8**	41.8	53.4
✓		**76.6**	38.6	60.3	58.5
	✓	76.5	**41.8**	42.1	53.5
✓	✓	**76.6**	38.6	**61.2**	**58.8**

the ProjRefine module enables the refinement of the proposal boxes generated by the first-stage RPN. However, for small-sized ship targets, the noise present in the weakly supervised segmentation masks may lead to false alarms. Since the M&S processing is only applied to large-sized slender targets, it has no impact on aircraft and ships. After M&S, the detected slender objects are more complete, and there are fewer false alarms.

Finally, to further investigate the impact of feature fusion strategies in the SA-DeRoI module, we compare four representative approaches: concatenation, weighted summation, channel attention (CA), and gating mechanisms. Let $F \in \mathbb{R}^{C\times H\times W}$ denote the original RoI-aligned features and $F_s \in \mathbb{R}^{C\times H\times W}$ the structure-aware features generated by the structural encoder. The fused feature F_c is computed as follows.

1) *Concatenation (Ours)*

$$F_c = \text{concate}(F, F_s). \tag{15}$$

For details, refer to (9). The fusion strategy preserves full information from both sources and allows the subsequent offset predictor to learn adaptive interactions.

2) *Weighted Summation*

$$F_{\text{fused}} = \alpha F + (1-\alpha)F_s \tag{16}$$

where $\alpha \in [0,1]$ is a learnable scalar weight shared across all channels.

3) *Channel Attention (CA)*

$$F_{\text{fused}} = W_c \odot F + (1-W_c)\odot F_s \tag{17}$$

where $W_c = \sigma(W_1[\text{GAP}(F);\text{GAP}(S)])$ denotes the channel-wise attention weights, GAP$(\cdot)$ is global average pooling, W_1 is a learnable weight matrix, σ is the sigmoid function, $[\cdot;\cdot]$ denotes channel-wise concatenation, and $\odot$ denotes channel-wise multiplication.

4) *Gating Mechanism*

$$G = \sigma(W_2[F; F_s]) \tag{18}$$

$$F_c = G \odot F + (1-G)\odot F_s \tag{19}$$

where W_2 is a 1×1 convolutional layer that outputs a gating map $G \in \mathbb{R}^{C\times H\times W}$.

These formulations represent a spectrum of fusion complexities, from parameter-free (concatenation) to adaptive, content-aware weighting. As shown in Table IX, simple concatenation yields the best overall performance, suggesting that explicit preservation of both feature streams benefits offset learning more than implicit reweighting.

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TABLE IX
ABLATION STUDY ON FEATURE FUSION STRATEGIES IN THE SA-DEROI MODULE

	Aircraft	Ship	Train	mAP (%)
Concat (Ours)	**76.6**	38.6	**61.2**	**58.8**
Weighted Sum	76.3	39.2	52.7	56.1
CA	76.2	**40.4**	53.8	56.8
Gate	76.5	40.0	56.1	57.5

TABLE X
DETECTION PERFORMANCE ON THE TRAINING CLASS UNDER VARYING IMAGE ROTATION ANGLES

Rotation Angle	Train mAP (%)
0°	**61.2**
15°	57.8
30°	57.3
45°	55.5
60°	54.2
90°	55.2

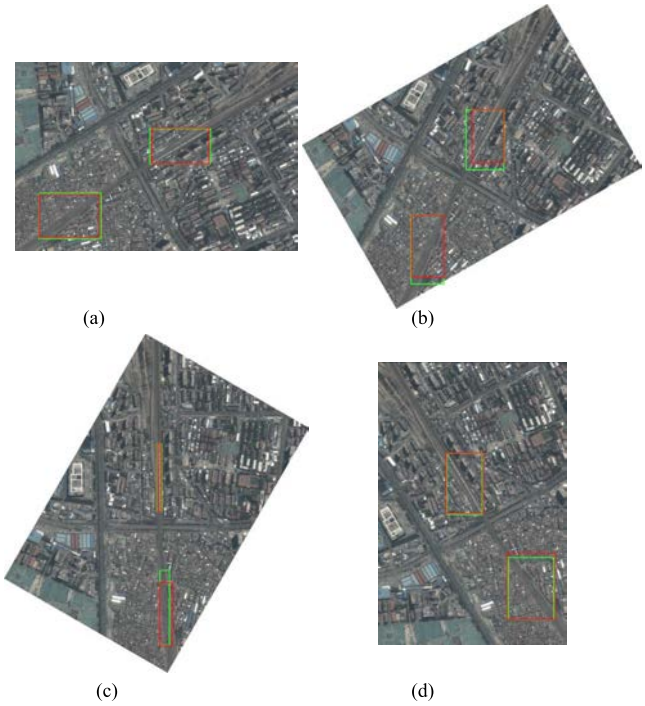


Fig. 11. Visualization of detection results on 30°, 60°, and 90° rotated training instances. (a) Original image. (b) Rotated 30°. (c) Rotated 60°. (d) Rotated 90°.

G. Robustness to Object Rotation

To evaluate the robustness of our geometric parallelism constraint under nonaxis-aligned orientations, particularly for slender objects, we conduct a controlled rotation experiment on the training class. Test images are rotated by fixed angles (0°, 15°, 30°, 45°, 60°, and 90°).

As reported in Table X, detection performance gradually declines with increasing rotation angle. This trend is primarily due to the boundary artifacts introduced during image rotation. To fill the empty regions created by rotation, we adopt edge reflection padding. Such padding introduces unnatural intensity transitions near image boundaries, and as the rotation angle increases, the extent of padded regions grows, leading to more pronounced artifacts that can interfere with feature representation and localization. Consequently, detection accuracy is slightly compromised.

Nevertheless, our method achieves 55.5% mAP at 45° and 54.2% at 60°, only 6% below the 0° baseline (61.2%). The slight recovery at 90° (55.2%) further indicates that the two-directional parallelism regularization effectively enforces geometric consistency between predicted bounding box edges and the underlying object structure, mitigating the adverse impact of boundary artifacts.

Qualitative results in Fig. 11 show that our detector consistently produces well-aligned bounding boxes even under large rotations, confirming the empirical robustness of our approach to arbitrary object orientations.

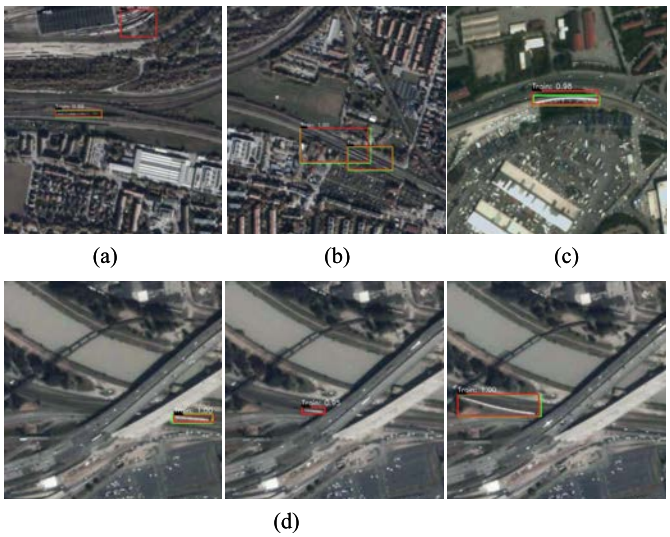


Fig. 12. Visualization for extreme situations. (a) Indistinguishable. (b) Close together. (c) Axis-parallel. (d) Occlusion.

V. CONCLUSION

The slender object exhibits a significant dimension along its central axial direction, while the dimension perpendicular to this axis is extremely small. Therefore, in contrast to the multiscale problems frequently studied in remote sensing, it is essential to simultaneously leverage both high- and low-level features of a target. In this article, we propose a geometric parallelism embedded structure-aware slender object detection method. This method employs the geometric parallelism regularized level set objective function as weak supervision to extract structure-aware features pertinent to slender objects. To mitigate the adverse influence of directly fusing low-level features on high-level features, structure-aware features are modified to learn sampling offsets of the SA-DeRoI to extract more relevant features related to slender objects in the box prediction branch. Furthermore, ProjRefine and M&S for fragments are designed for large-scene and large-size slender object detection to reduce the false alarms. The experimental results show that GPSDet outperforms other methods in terms of quantitative metrics and qualitative results on two public datasets.

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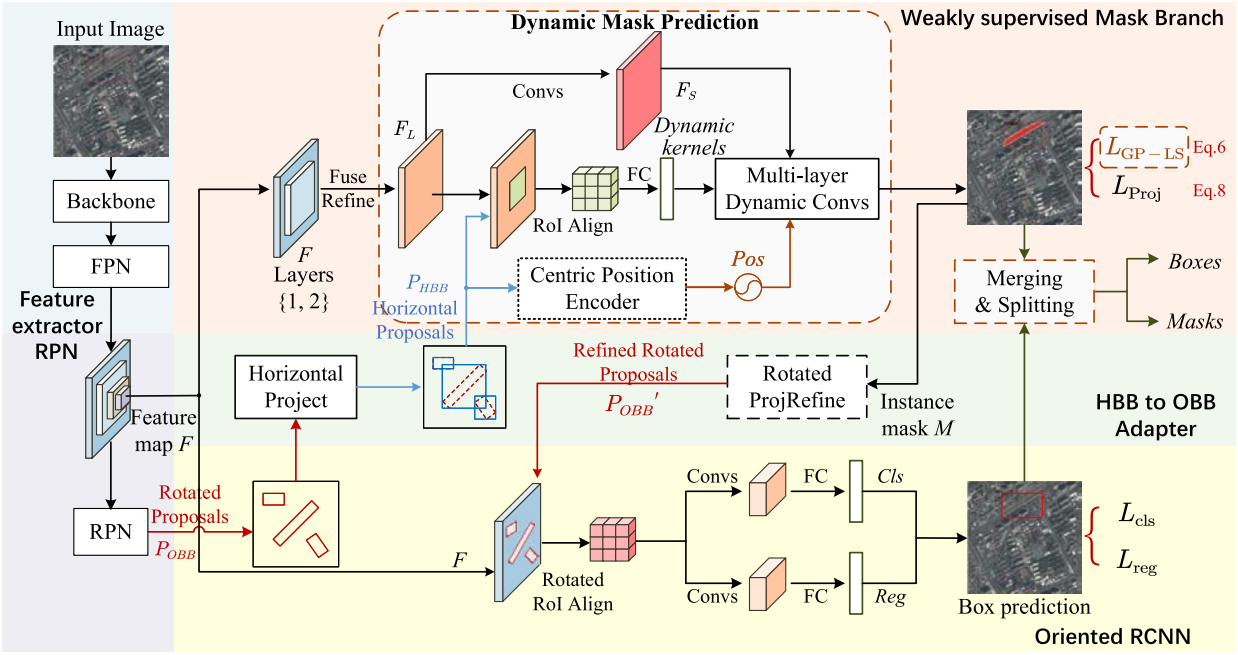


Fig. 13. Structure for Rotated-GPSDet.

However, several directions for future work remain. SAT-MTB is a multitask satellite video dataset, where numerous samples originate from the same video sequence, impacting the diversity of the training set. On the other hand, multiple instances of airplanes and ships are frequently present within the same scene, while the number of trains is relatively scarce. This results in a class imbalance within the dataset, which subsequently affects the stability of training and the robustness of the model. However, advantageously, trains exhibit smaller displacement compared to their length and possess uniform trajectories, leading to a large overlap of targets across adjacent frames. Leveraging this characteristic to explore the utilization of temporal information in the detection of slender objects represents a promising direction for future research.

APPENDIX
VISUALIZATION FOR EXTREME SITUATIONS

From the extended SAT-MTB+ dataset, we selected representative challenging cases for visualization in Fig. 12. Here, green boxes denote the ground truth, while red boxes represent the detection results from GPSDet. The proposed method demonstrates robust detection for axis-parallel instances and can even detect samples with low contrast against the background. However, in cases such as container clusters, which share similar visual features with trains, the detector may be misled, resulting in false alarms.

A. Rotated-GPSDet

Similar to HBB approaches, oriented object detection can generally be categorized into single-stage and two-stage frameworks. Conventional single-stage rotated object detectors struggle to handle large aspect ratio variations among intraclass targets, particularly in remote sensing scenarios where objects may appear in dramatically different orientations and proportions. The two-stage framework addresses this through predicting rotated region proposals in the first stage and then extracting more target-aligned regional features in the box head. This architecture mitigates the "small target area" problem inherent in horizontal proposal boxes, as the rotated proposals better encompass oriented objects. However, the current two-stage rotated detectors still rely on proposal area for feature hierarchy division, inheriting the fundamental limitation of lacking detailed target information, which is necessary for slender object detection.

The proposed weakly supervised mask head can be effectively transferred to rotated object detection frameworks. Taking Oriented R-CNN as an example, the architecture of Rotated-GPSDet is illustrated in Fig. 13. To adapt to the rotated bounding box detection framework, we introduce an additional adapter module. The horizontal projection component converts the rotated proposal boxes into horizontal boxes as input for the mask branch, while the rotated projection refinement module maps the predicted instance masks back to rotated bounding boxes.

Experimental results on the GLH-Bridge dataset demonstrate the effectiveness of this approach, as shown in Table XI.

TABLE XI
ROTATED OBJECT DETECTION EXPERIMENTS ON THE GLH-BRIDGE DATASET

	$AP_{50:95}$	AP_{50}	AP_{75}	AP_{sh}	AP_{md}	AP_{lg}	AP_{hg}
Oriented RCNN	30.8	**66.4**	23.8	29.5	33.8	21.0	3.1
w/o. ProjRefine	32.0	66.0	24.6	29.9	**35.3**	19.0	5.1
w/o. M&S	31.7	65.3	24.6	**30.7**	35.0	23.2	8.4
Rotated-GPSDet	**32.5**	**66.4**	**25.3**	30.6	33.4	**30.0**	**17.6**

TABLE XII
ROTATED OBJECT DETECTION EXPERIMENTS ON THE DOTA DATASET

	$AP_{50:95}$	AP_{50}	AP_{75}	AP_{sh}	AP_{md}	AP_{lg}	AP_{hg}
Oriented RCNN	24.8	37.6	**14.2**	26.6	27.4	**22.8**	0.0
Rotated-GPSDet (w/o. M&S)	**25.7**	**44.3**	14.1	**27.5**	**28.3**	22.7	0.0

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The experimental results demonstrate that the proposed weakly supervised mask branch can also enhance the detection performance of rotating bounding boxes for slender objects. We also evaluated our method on the bridge category of the DOTA-v1.0 dataset, with the results presented in Table XII. However, according to the GLH-Bridge definition of huge bridges, there are only 5 instances in the DOTA-v1.0 test set. Consequently, the performance gap between our proposed method and the baseline (Oriented RCNN) is not significant.

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